



Jargon Buster

Dots per inch (dpi): Image resolution in either the vertical or horizontal axis. On an average, printers have a resolution of 600 dpi vertical and 600 dpi horizontal.

Duplexing: A printer's ability to print on both sides of a page in one pass.

Duty cycle: The number of pages a printer can produce per month.

Nozzle (or jet): Tiny tubes, which allow for a heated ink bubble to travel through as it cools. This is the one vital area of the print head which needs to be cleaned occasionally or replaced.

Piezo technology: Epson's proprietary inkjet technology uses a piezo crystal at the back of the ink reservoir. This is rather like a loudspeaker cone—it flexes when an electric current flows through it and in so doing forces a drop of ink out of the nozzle.

Print alignment: The proper setting of the ink nozzles and the print head.

Print head: An important part of a printer, it contains electrical circuitry and the nozzles to heat and disperse the ink during printing. You should avoid touching or cleaning the print head unless you have a manufacturer-approved solution.

Printer memory/RAM/input buffer:

Memory inside your printer that stores data sent from your computer. The more complex your documents, the more printer memory you need. It is also known as the storage memory your printer uses to hold incoming data when it arrives faster than your printer's engine can handle. This results in less waiting time.

Print speed: This is measured in pages per minute (ppm) through either of two methods—'rated engine speed' or the number

of pages of text that a printer can produce in one minute using internal fonts, and 'throughput' or how fast raw data is turned into hard copy.

Resolution: The image quality is usually measured by the number of dots per inch (dpi) printed. When describing halftone image quality, it may be measured by lines per inch. The more dots or lines per inch, the better the image quality.

Thermal technology: Most inkjets use thermal technology, whereby heat is used to fire ink onto the paper. The squirt is initiated by heating the ink to create a bubble until the pressure forces it to burst and hit the paper. The bubble then collapses as the element cools, and the resulting vacuum draws ink from the reservoir to replace the ink that was ejected. This is the method favoured by Canon and Hewlett-Packard.

Lexmark has done a very good job with its drivers. The interface is easy to work with and there are on-screen ink level indicators. All Lexmark printers include a data cable. HP and Olivetti too provide data cables. Canon on the other hand does not provide a data cable with the S200SP and the BJC 2100SP.

In the mid-range category, almost all printers provide a minimum of 1200x1200 dpi printing. All three HP models in this category, the 845C, the 920C and the 948C, offer USB as well as parallel port support and also provide a data cable and come with an impressive 2 MB buffer.

The Canon S300, which had the highest overall score in this category, lets you check ink level status through the driver. It also has built-in indicators for low ink and out-of-paper, but no indicator to alert for a paper jam. It supports dual-sided printing, but only offers a USB interface, which makes it incompatible with older PCs. Also, no data cable is included and it sports only 56 KB of input buffer.

Though the Olivetti Artjet 22 logged the lowest overall score in the mid-range category, it sports some very interesting features. It can be set up horizontally or vertically, which makes it quite versatile if space is an issue. It includes parallel and USB interfaces and supports a paper capacity of 150 sheets, so bulk printing should not be a problem.

Among the high-end printers, the Epson Stylus 880 scored the highest for features. It supports a resolution of 2880x720 and provides both USB and parallel connectivity. It has support for dual-sided printing, shows accurate status of ink tanks through the driver and to top it all, supports iron-on transfers and transparency printing! This model also had LED indicators to alert for paper jams, low ink levels and for when the printer is out of paper. The other entries from Epson in this category (Stylus Photo 810, Stylus 895 and Stylus 1520) also included such LED indicators. All supported USB as well as parallel connectivity, except the Stylus 1520, which offers a parallel and a Mac Serial interface.

While all printers in this category support A4 printing, the Epson Stylus 1520, the HP Deskjet 1125C and the Canon S4500 have support for printing pages in A3 size as well.

The Canon S4500 occupies a lot of desktop space due to its large footprint. It supports a maximum resolution of 1440x720 and comes with 64 KB onboard buffer. It can communicate with the PC through the USB or the parallel interface. There is no data cable provided, nor does it include any printing media. This printer too can be configured as a scanner. All one has to do is replace the ink cartridge with the IS-32 scanner cartridge to scan at 720 dpi.

The Lexmark Z65N too does not fail to impress. It has a mammoth resolution of 4800x1200 dpi and has a huge paper capacity of 250 pages. But the feature that takes the cake is the fact that it comes with integrated network support. Also, paper can be fed into the printer through two separate trays.

Performance

A printer's performance is judged by the quality of output and the speed at which it is capable of printing. Whether you should consider speed or quality of output when buying a printer will depend on how you plan to use the printer. If you need to print in large volumes, then speed should rate higher in your scheme of things. But if you need prints for presentations, or prints of photographs, then quality of output (along with speed) is critical. In fact, if a printer gives you great quality, then you can consider compromising a little on the speed front.

Speed: The speed limitation comes from a variety of factors. The first is what is called the cycle time or the pulse cycle. This is the time taken by the printer head to print one line and then move on to the next. Obviously, a shorter cycle time means higher printing speed. The second factor that affects printing speed is the amount of input buffer available. A large buffer will allow you to print large documents faster, and will result in shorter spool time if the printer is shared over a network.

Entry-level printers usually compromise on speed in favour of quality, as a typical budget user wants the perfect balance of quality and speed. Here, a printer that can print a plain black and white sheet under 30 seconds is just about fine. On an average, most printers in the budget range managed to print the black and